Final Year Research Project Proposal

on

Water quality analysis health risk assessment from fluoride and ammonia

by

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1 Project Overview

The goal of this project is to develop a water-quality prediction system that can forecast the current and future risk level of ammonia and fluoride in drinking water. Water safety and human health are directly impacted by these two factors. Based on past trends, the project will also predict future changes in fluoride and ammonia levels. In order to achieve this, we will build and train three deep learning models—LSTM, GRU, and Transformer and compare their performance. The main objective is to help detect unsafe water conditions early and support better water-treatment decisions.

2 Background and Motivation

Ammonia and fluoride are key parameters for checking water quality. High fluoride levels can cause Dental fluorosis (occurs when children are exposed to high fluoride levels during tooth development, leading to white streaks on enamel, though tooth health remains unaffected.) Skeletal fluorosis (Develop after long-term high exposure, causing bone and joint pain, reduced elasticity, and increased fracture risk.) Similarly High ammonia levels can affect the liver and kidneys because the body must work harder to detoxify ammonia. It can also react with chlorine during water treatment to form chloramines, which may cause respiratory irritation.

Regular water testing is not available in many places, and dangerous levels are frequently discovered too late. With the development of data-driven technologies, deep learning models can assist in anticipating risky trends before they become serious.

This motivates our project, which aims to develop a forecasting and prediction system that analyses past water-quality trends/patterns to safeguard communities and enhance water management planning.

3 Objectives

The key objectives of the project are

1. To collect, analyze, and preprocess water quality data.
2. To build three deep learning forecasting models (LSTM, GRU, and Transformer) that can classify the current risk level of ammonia and fluoride in water and to predict the future levels of fluoride and ammonia.
3. To compare the performance of the three models and identify which one works best for long-term forecasting.
4. To visualize trends and help authorities understand potential risks early.

4 Proposed Methodology

- Data Collection: Gathering the dataset from Telangana Pollution Control Board, which should contain two chemical which is ammonia and fluoride.
- Data Preprocessing: Cleaning the data by handling missing values and removing unwanted data. Normalize the data.
- Feature Engineering: Extract useful patterns from the data and create input sequences for LSTM, GRU and Transformer models.
- Model Development, Training and Validation: Build three forecasting models: LSTM, GRU, and Transformer. Split data into training and testing sets. Train each model and fine-tune parameters to improve accuracy.Validate performance using metrics like RMSE and accuracy curves.
- Model Comparison: Compare the performance of LSTM, GRU, and Transformer models.Identify the best model for accurate long-term forecasting.
- Visualization and Interpretation: Use graphs to show trends in fluoride and ammonia levels over time.
- System Development: Build a simple UI that displays Current risk level, Predicted future values,Trend chart.

5 Tools to be Used

Provide the list and description of tools (Software, Platform, Equipments etc.) to be used in the completion of the Project.

- **Programming Languages: Python**
- **Frameworks & Libraries: Pandas, Numpy, Sklearn, Tensorflow, Streamlit**
- **Dataset: Water Quality Data by Telangana Pollution Control Board**
- **Development Platform: Juypter Notebook, Google Colab**
- **Visualization Tools: Matplotlib, Seaborn**

6 Expected Outcomes

The expected deliverables of the project are:

- Visualisations showing the change in fluoride and ammonia concentrations over the monitoring period. Used to identify significant trends.

7 Timeline

Project Phase	Duration
Literature Review and Dataset Collection	Feb 1 – Feb 22
Data Preprocessing and Feature Engineering	Feb 23 – March 15
Model Development and Training	March 16 – April 12
Recommendation System and UI Development	April 13 – May 3
System Integration and Testing	May 4 – May 17
Documentation and Final Submission	May 18 – May 24

8 Team Roles

- **Aditya Ranjan Prusty: Dataset Collection, EDA**
- **Adviteeya Pattanaik: Preprocessing, Model Development**
- **Jagadeesh Pradhan: Model hyper-parameter tuning. UI/UX development, System intergration, deployment**
- **All Members: Documentation, presentation, testing, and debugging.**

9 Contact Details of Team Lead

- **Name: Jagadeesh Pradhan**
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- **Mobile No: 8460734722**

10 References

- MiaomiaoTian, Wenzhao Li*, MeijuanRuanJingWei & WeiweiMa (2019). *Water Quality Pollutants and Health Risk Assessment for Four Different Drinking Water Sources.*
- Mahmood Yousefi, Mahboobeh Ghoochani & Amir Hossein Mahvi (2017). Health risk assessment to fluoride in drinking water of rural residents living in the Poldasht city, Northwest of Iran.
- Dataset: `https://tinyurl.com/3vecyder`

Student Signatures

We hereby declare that the above project proposal is prepared by us and all information provided is true to the best of our knowledge.

______________________	______________________
(Student 1 Name)	**(Student 2 Name)**
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(Student 3 Name)	**(Student 4 Name)**

(Office use only)

Assigned Supervisor: ______________________.

Comments

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Project Status

Approved	**Not Approved**

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(Dr. Rangaballav Pradhan)	**(Gyana Ranjan Patra)**

(Project Coordinators)

Date:______________________ .

Place:______________________ .